FIBONACCI SEQUENCE

Binet's Formula

An explicit formula for a recursively defined sequence.

THEOREM

$$F_n = \frac{\phi^n - \psi^n}{\sqrt{5}} \text{ with } \phi = \frac{1+\sqrt{5}}{2} \text{ and } \psi = \frac{1-\sqrt{5}}{2}.$$

1 GUESS A POWER

Assume $F_n = x^n$ in $F_n = F_{n-1} + F_{n-2}$: $x^n = x^{n-1} + x^{n-2}$.

2 CHARACTERISTIC EQUATION

$x^2 = x + 1$, with roots ϕ and ψ.

3 COMBINE

The recurrence is linear, so $F_n = A\phi^n + B\psi^n$; $F_1 = F_2 = 1$ fix A and B.

4 READ THE GROWTH

$|\psi| < 1$, so ψ^n vanishes for large n and $F_n \approx \phi^n/\sqrt{5}$: about 61.8% growth per step.

DROPPING THE SIGN OF THE SECOND ROOT

Using $\psi \approx +0.618$ gives $F_3 \approx \frac{4.236 - 0.236}{\sqrt{5}} \approx 1.789$, not a whole number.

With $\psi \approx -0.618$, $F_3 = \frac{4.236 + 0.236}{\sqrt{5}} = 2$.